EXPANDING CONFUSION

For the last several years, measurements of the Hubble Constant from supernovae have been giving us a number around 74 km/s/Mpc—that means that a galaxy one megaparsec away (that's around 3.2 million light-years) is receding from us at around 74 km/s. One twice as far away is moving, relative to us, about twice as fast. But we can also measure the Hubble Constant indirectly, by carefully studying the geometry of the hot and cold spots in the cosmic microwave background. When we measure it that way, the number we get is closer to 67 km/s/Mpc. Even though these observations are looking at very different epochs of cosmic history, each of them can tell us the expansion rate today. In a universe made of what we think it's made of, both methods of determining the Hubble Constant really ought to give us the same number. And they don't.

This hasn't always been considered to be *that* big a problem, since no one thought either measurement was so incredibly precise as to settle the question. Until recently, the state of play was that the cosmic microwave background folk assumed that there was some distance ladder mis-estimate that would be sorted out eventually, dropping the number down a tad, and the supernova folk figured that the CMB measurements, which derive ultimately from attempting to measure the shape of space itself, were so complicated that surely something would show that the number was really just a little bit higher. This isn't an unreasonable assumption, given the number of calculations and conversions that go into looking at a baby picture of the universe and converting that into a present-day expansion rate. And the distance ladder, likewise, really is fantastically complicated. Before even getting into all the possible biases that might creep in if you don't account for every relevant property of the supernovae themselves, calibrating variable stars is not easy, and even distances to relatively nearby galaxies sometimes come with huge uncertainties. Part of this is due to how the populations of Cepheid variables we can see nearby are different from those far away, and … well, I could go on. Let me just say there are *debates*.

While the assumptions from each side that the other has done something wrong haven't quite gone away, the situation is getting increasingly uncomfortable due to the fact that both sides are improving their methods,